

Problem Set 3

Fundamentals of Numerical Programming I

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Problem 1. Numerical derivatives by finite differencing

Consider a continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$. Based on evaluations at points x_i with $x_{i+1} = x_i + \Delta x$, we would like to approximate the derivative $f'_i \equiv \frac{df}{dx}(x_i)$.

In the lecture, we have derived

- the first order forward difference formula

$$f'_i = \frac{f_{i+1} - f_i}{\Delta x} + \mathcal{O}(\Delta x) \quad (1)$$

- and the second order central difference formula

$$f'_i = \frac{f_{i+1} - f_{i-1}}{2\Delta x} + \mathcal{O}(\Delta x^2) \quad (2)$$

based on Taylor expansion.

- (a) We would now like to derive a symmetric 4-th order finite difference formula to approximate f'_i . Determine the coefficients a and b in the symmetric Ansatz

$$f'(x) = a(f(x + \Delta x) - f(x - \Delta x)) + b(f(x + 2\Delta x) - 2f(x - \Delta x)) + \mathcal{O}(\Delta x^4) \quad (3)$$

to obtain a 4-th order finite difference approximation of $f'(x)$ by Taylor expansion.

- (b) Implement Python functions to calculate

- first order forward differencing formula
- second order central differencing formula
- symmetric fourth order differencing formula

You may pass the following inputs:

- the function to be differentiated
- the evaluation point
- Δx

We will call the result from the implementation of the finite difference formula $\hat{f}'_i$ from now on.

- (c) Consider the function $f(x) = L(L(L(x)))$ where L is the *logistic map* $L(x) = 4x(1-x)$.
- Find the analytical derivative of f by chain rule. *Hint:* You don't need to carry out the multiplications.
 - For $x_0 = 0.2$ and $\Delta x = \text{np.logspace}(-15, -3, \text{num} = 200)$ calculate the error term $\text{err} = |\hat{f}'_0 - f'_0|$ for all three finite difference formulas. Plot the error term over Δx in a double-logarithmic plot.
- (d) Where is the round-off error, where the truncation error dominant? Explain the behavior you observe.

Problem 2. Newton interpolation and Runge's phenomenon

Consider the *Runge* function

$$f_A(x) = \frac{1}{1 + 25x^2} \quad (4)$$

as well as

$$f_B(x) = \sin(x) \quad (5)$$

and

- equidistant evaluation points

$$x_i = \frac{2i}{m} - 1, \quad i = 0, \dots, m \quad (6)$$

- as well as *Chebyshev-nodes*

$$\tilde{x}_i = \cos\left(\frac{(2i+1)\pi}{2(m+1)}\right), \quad i = 0, \dots, m \quad (7)$$

In this task, we will perform **Newton interpolation**

$$p(x) = \sum_{j=0}^m d_j n_j(x), \quad n_j(x) = \prod_{i=0}^{j-1} (x - x_i) \quad (8)$$

with the coefficients given by the finite differences

$$d_j = [x_0, \dots, x_j]f \quad (9)$$

where for $j < i$ we recursively define

$$\begin{aligned} [x_j]f &= f_j, \\ [x_j, \dots, x_i]f &= \frac{[x_{j+1}, \dots, x_i]f - [x_j, \dots, x_{i-1}]f}{x_i - x_j} \end{aligned} \quad (10)$$

- Implement Newton divided differences in Python.
- Implement the modified Horner scheme to evaluate $p(x)$ via

$$z_m = d_m, z_j = d_j + (x - x_j)z_{j+1} \quad \rightarrow \quad z_0 = p(x) \quad (11)$$

- For both f_A and f_B perform Newton interpolation for $m = 3, 9, 24, 48$ both with

equidistant and Chebyshev nodes. Plot the results. For plotting $p(x)$ use $m = 200$ evaluation points. Interpret your results.

Problem 3. *(Optional)* The power of finite differencing - solving the Kuramoto–Sivashinsky equation

The Kuramoto–Sivashinsky equation¹ is given by

$$\partial_t u + \alpha L \Delta u + \beta L^3 \Delta^2 u + \frac{\gamma L}{2u_s} |\nabla u|^2 = 0 \quad (12)$$

where Δ is the Laplace and Δ^2 the biharmonic operator. $L = 1.0$ is the box size and α, β, γ are parameters, here

$$\alpha = 10^{-2}, \quad \beta = 10^{-6}, \quad \gamma = 10^{-2} \quad (13)$$

(physically all are in velocity units) and $u_s = 1.0$.

We consider a 2D setting, so

$$\Delta = \partial_x^2 + \partial_y^2 \quad (14)$$

$$\Delta^2 = (\partial_x^2 + \partial_y^2)(\partial_x^2 + \partial_y^2) = \partial_x^4 + \partial_y^4 + 2\partial_x^2 \partial_y^2 \quad (15)$$

We assume periodic boundary conditions and a box of size $L = 1$.

We discretize space to a grid with N^2 grid points

$$\begin{aligned} x_i &= L \frac{i}{N}, \quad i = 0, \dots, N-1 \\ y_j &= L \frac{j}{N}, \quad j = 0, \dots, N-1 \end{aligned} \quad (16)$$

such that the grid-spacing is $\Delta x = \frac{L}{N}$.

Let us define the distance to the center of the box as

$$r_{ij} = \sqrt{\left(x_i - \frac{L}{2}\right)^2 + \left(y_j - \frac{L}{2}\right)^2} \quad (17)$$

The initial conditions are given by

¹Here rather a physicists version of it.

$$u_{ij}(t=0) = \exp\left(-\eta \frac{r_{ij}}{L}\right), \quad \eta = 500 \quad (18)$$

Time is discretized into an iteration with N_t steps of size Δt with

$$t_{\text{final}} = 1.5L, \quad \Delta t = \frac{10^4}{L^3} \Delta x^4, \quad N_t = \lfloor \frac{t_{\text{final}}}{\Delta t} \rfloor \quad (19)$$

- (a) Implement discrete Laplace, Biharmonic and gradient-squared operators based on the first and second derivative implementations given in Code-Snippet 1.
- (b) Implement the right-hand side of the KS equation

$$\partial_t u = -\alpha L \Delta u - \beta L^3 \Delta^2 u - \frac{\gamma L}{2u_s} |\nabla u|^2 \quad (20)$$

Use the discrete Laplace, Biharmonic and gradient-squared operators just implemented.

- (c) Implement a time integration loop taking in an initial state the time step Δt and the number of time steps N_s and returning the final state. A single time step is given by

$$u(t + \Delta t) \approx u(t) + \Delta t \partial_t u(t) \quad (21)$$

$\partial_t u(t)$ given by the function you have just implemented.

- (d) Run the simulation with the given initial conditions. Plot the final state using `plt.imshow`. The expected result is shown in Fig. 1
- (e) Produce an animation of the state over time. Do not use every physical time step of your simulation as a frame but rather 150 frames equally spaced in time.

```

1      import numpy as np
2
3      XAXIS = 0
4      YAXIS = 1
5
6      def first_deriv(u, grid_spacing, axis):
7          return (np.roll(u, 1, axis=axis) - np.roll(u, -1,
8                  ↪ axis=axis)) / (2 * grid_spacing)
9
10     def second_deriv(u, grid_spacing, axis):
11         return (np.roll(u, 1, axis=axis) - 2 * u + np.roll(u, -1,
12                 ↪ axis=axis)) / grid_spacing**2

```

Code-Snippet 1: Finite difference approximations of the first and second derivative along a specified axis of an array. By rolling the arrays, we implicitly assume periodic boundary conditions.

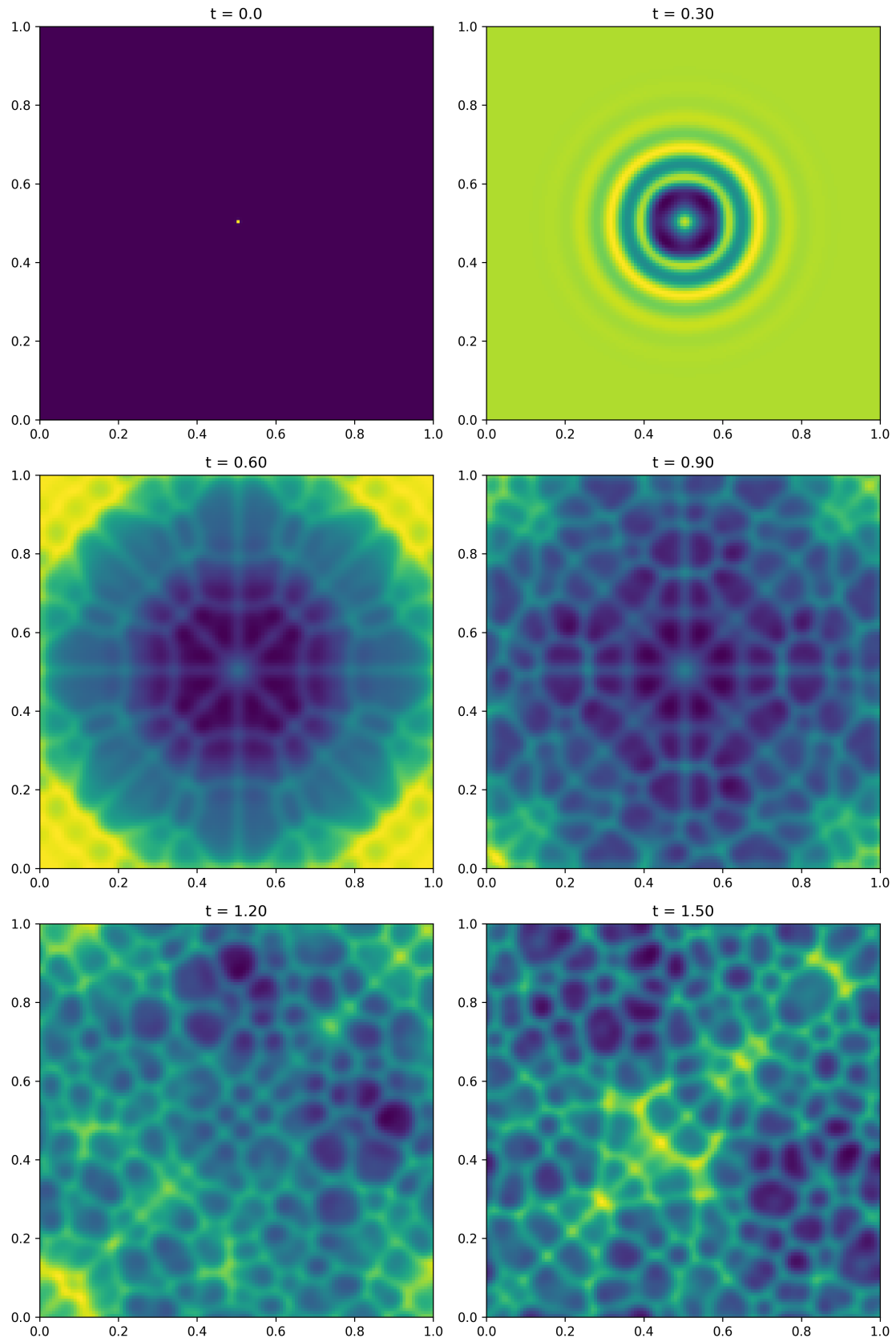


Figure 1: States in the time evolution of the Kuramoto–Sivashinsky equation.